Secondary treatment

Secondary treatment removes more than 85 percent of both suspended solids and BOD. Secondary treatment involves the oxidation of the primary effluent by microorganisms. The common types processes used are:

1. Trickling filter process
2. Activated Sludge process
3. Oxidation ponds

A trickling filter consists of a large tank or basin filled with a bed of crushed stone, gravel, slag or other porous material. Sewage is sprayed in a fine mist over the rocks. As the sewage trickles through the bed, organic matter clings to the rocks, where it is digested by heterotrophic microorganisms (Figure 9.9). The microorganisms are contained within a biofilm which are produced by slime forming bacteria such as Zooglea and the organic matter, is oxidized to gases and inorganic products.

In the activated sludge process sewage is mixed with a slime forming bacteria (Zooglea) in a large aeration tank. As the mixture is aerated, large flocs, or clumps, form. These clumps contain not only the original slime forming bacteria but also large population of heterotrophs, which oxidize organic matter within these clumps. In this system wastewater is continuously pumped into the tank and the treated water is removed into a holding tank and the flocs are allowed to settle. The settled floc material is recycled into the tank as an inoculums to continue the process. The remaining floc is further treated or removed for burial or incineration.

Oxidation ponds (Figure 9.10) are used in some communities to handle small loads of sewage. Small communities and isolated areas frequently use oxidation ponds for treatment of waste water. Sewage is channeled into an initial pond where the sludge settles out. The liquid portion of the sewage is then pumped into an adjacent series of ponds where aeration allows bacterial growth and degradation of organic matter. These secondary ponds often are seeded with algae which provide oxygen for the growth of aerobic, heterotrophic bacteria.

Tertiary treatment

Tertiary processes can remove 99 percent of all the impurities from sewage. Although effluents from secondary treatment have a low BOD, they may contain eutrophication inducing salts (Phosphorus and Nitrogen compounds), organic and inorganic suspended solids, and poorly biodegradable organic materials. Advanced or tertiary treatment process are designed to reduce or eliminate these materials depends more on physical and chemical processes than biological processes. For phosphorus elimination the phosphates are converted to poorly soluble aluminum, calcium or iron compounds and removed by precipitation. Nitrogen in sewage effluent is removed primarily through nitrification by microorganisms. The extent of nitrification during tertiary treatment depends on adequate treatment plant designed and the proper removal of sludge so that these bacteria are grown under optimal conditions. Otherwise, large amounts of nitrogenous compounds may escape tertiary treatment and released in the effluent. Suspended

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